

6 (**long**) A circular, straight pipe enclosing air of density ρ undergoes a smooth contraction from diameter D_1 to diameter D_2 . A series of pressure taps are installed along the pipe and connected to manometers containing a column of liquid of density ρ_ℓ , as shown in Fig. 3 when the air is at rest. The air is subsequently made to flow such that upstream of the contraction it has velocity V_1 and pressure p_1 . Upstream of the contraction, pressure tap 1 is installed with the tube opening at a right angle to the flow direction. Pressure tap 2 has the opening facing the flow and is connected to a Pitot tube. Downstream of the contraction, pressure taps 3 and 4 are set in a similar arrangement, as shown in the Figure. The manometers are labelled with two digits marking the taps that they connect, e.g. manometer 24 connects taps 2 and 4, and they give as a reading the height difference in their column of water, e.g. h_{24} , with the reading being positive when the right side is higher. The manometers are deep enough not to overflow. Friction forces may be neglected throughout. The effect of gravity g on the air flow may also be neglected.

- (a) Find the velocity downstream of the contraction, V_2 , in terms of V_1 , D_1 and D_2 . [2]
- (b) Find the pressure downstream of the contraction, p_2 , in terms of p_1 , ρ , V_1 , D_1 and D_2 . [4]
- (c) For all the manometers, explain which readings you expect to be zero, and which ones equal and which ones opposite to each other. [14]
- (d) Give in terms of ρ , ρ_ℓ , g , V_1 , D_1 and D_2 :
 - (i) the reading in manometer 12, h_{12} . [2]
 - (ii) the reading in manometer 13, h_{13} . [4]
 - (iii) the reading in manometer 23, h_{23} . [4]

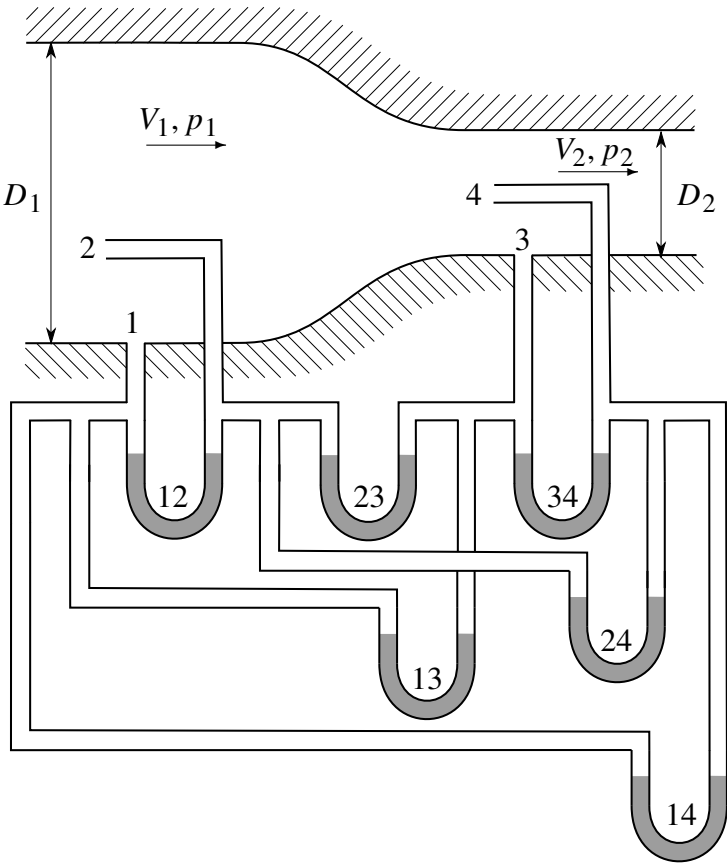


Fig. 3